ChatGPT - Homework

L18.pdf
PDF

You are given lecture slides and relevant textbook excerpts as the context. Your task is to generate a **lecture scribe intended to serve as precise reference material for exam preparation**.

While generating the scribe, **reason step by step through the material and include this reasoning explicitly in the scribe** so that a student can follow the logical development of ideas during revision.

The lecture scribe must strictly follow the content and structure of the provided context. **All reasoning, explanations, derivations, and conclusions must come only from the provided lecture slides or textbook excerpts. Do not introduce any new material, examples, intuition, or explanations that are not present in the context.**

When writing the scribe:

• Present **definitions, notation, assumptions, and conditions explicitly**.
• Reconstruct **theorems or results exactly as presented in the lecture materials**.
• Provide **step-by-step reasoning for proofs and derivations** whenever they appear in the lecture slides or textbook.
• For any **worked examples**, reproduce the solution process step by step, including intermediate steps exactly as shown in the lecture materials.
• Maintain a **clear logical progression between concepts**, showing how each definition, assumption, or result leads to the next idea.

The final output should:

• Be written in a **formal academic tone suitable for exam revision**.

• Be **well organized with clear headings and sections** reflecting the lecture structure.
• Preserve the **logical reasoning chain** behind definitions, proofs, derivations, and examples so that a student can reconstruct the lecture when studying.

Important constraints:

• Use **only the provided context** (lecture slides and relevant textbook content).
• **Do not invent content or fill gaps with external knowledge.**
• Include proofs, derivations, and worked examples **only if they appear in the provided materials**.

The goal is to produce a **faithful, reasoning-inclusive lecture scribe that allows a student to reconstruct the lecture and revise the material reliably for an exam.**

Below is a **faithful, structured lecture scribe** reconstructed strictly from the provided lecture slides. All definitions, reasoning steps, derivations, and examples are presented exactly as developed in the material, with explicit logical progression.

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

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1. Marginal PMF

1.1 Definition

For discrete random variables X and Y, the joint PMF is:

$$p_{XY}(x,y) = P(X = x, Y = y)$$

Derivation of Marginal PMFs

The marginal PMFs are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea (from slide)

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

1.2 Marginal PMF from a Joint PMF Table

From the joint PMF table:

- **Row sums** → $p_X(x)$
- **Column sums** → $p_Y(y)$

Formally:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Also:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

This establishes that marginal PMFs are valid probability distributions.

1.3 Solved Example

Given Joint PMF Table (page 5)

	$y = 0$	$y = 1$
$x = 0$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 1$	$\frac{3}{8}$	$\frac{1}{4}$

Step-by-Step Solution

Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Marginal PMFs

$$p_X(x) = \begin{cases} \frac{3}{8}, & x = 0 \\ \frac{5}{8}, & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} \frac{1}{2}, & y = 0 \\ \frac{1}{2}, & y = 1 \end{cases}$$

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2. Correlation

2.1 Definition

A basic way to measure joint behavior is through the **product moment**:

$$R_{XY} = E[XY]$$

Interpretation

- It gives the **average value of the product XY**
- It is referred to as a **raw correlation measure**

2.2 Limitations of R_{XY}

From the slides:

- Depends on the **scale of X and Y**
- Computed **about the origin**, not about the means
- Therefore, it mixes:
  - Mean values
  - Spread

- Dependence

Important Observation

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

2.3 Transition to Covariance

To remove the effect of means:

- **Center the variables first**
- This leads to the concept of **covariance**

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3. Covariance

3.1 Definition and Interpretation

Covariance is defined as:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Reasoning Behind Definition

1. Subtract the mean from each variable
2. Measure how the **centered quantities vary together**

Interpretation (from slides)

- Covariance captures whether deviations:
  - Occur in the **same direction** → Positive
  - Occur in **opposite directions** → Negative
  - Have **no linear relation** → Zero
- It indicates the **direction of joint linear variation**
- However:
  - Its magnitude depends on units
  - It **cannot measure strength of relationship**

3.2 Algebraic Form and Derivation

Start with:

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand the expression

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Substitute $\mu_X = E[X], \mu_Y = E[Y]$

$$\mathrm{Cov}(X, Y) = E[XY] - \mu_X\mu_Y$$

$$\boxed{\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

3.3 Properties of Covariance

1. **Symmetry**

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

2. **Variance as Special Case**

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

3. **Linearity with Constants**

$$\mathrm{Cov}(aX + b, cY + d) = ac \cdot \mathrm{Cov}(X, Y)$$

4. **Independence Implies Zero Covariance**

$$X \perp Y \Rightarrow \mathrm{Cov}(X, Y) = 0$$

Important Note

Zero covariance means **uncorrelated**, but **does not imply independence**.

3.4 Example 1

Problem

If:

$$Y = -2X + 3$$

Find:

$$\mathrm{Cov}(X, Y)$$

Step-by-Step Solution

Using:

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 1: Compute XY

$$XY = X(-2X + 3) = -2X^2 + 3X$$

Step 2: Compute expectation of Y

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step 3: Substitute into covariance formula

(As instructed in slides: substitute both expressions into covariance formula)

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

(Further steps follow substitution as indicated in continuation slides.)

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4. Pearson's Correlation Coefficient

Definition

- Measures the **effect of change in one variable when the other changes**

Properties

- Measures **strength and direction of linear relationship**
- **Normalized version of covariance**
- **Unit-free**

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5. Jointly Gaussian Random Variables

5.1 Definition

A pair (X, Y) is **jointly Gaussian** if:

- The joint density has the **bivariate Gaussian form**
- The **marginals are Gaussian**

5.2 Applications (from slides)

- Signal processing
- Communication systems
- Modeling continuous noisy measurements

5.3 Example (from slide)

Scenario

- X: Atmospheric CO_2 concentration
- Y: Global surface temperature anomaly

Because both vary due to many random processes, their joint behavior can often be approximated by a **joint Gaussian distribution**.

5.4 Algebraic Form and Interpretation

(Referenced in slides as continuation; algebraic form and interpretation follow from the definition of joint Gaussian density.)

Final Logical Flow Summary

1. Start with **joint PMF**
2. Derive **marginal PMFs** via summation
3. Measure joint behavior using $E[XY]$
4. Identify limitation → depends on means
5. Introduce **covariance** by centering variables
6. Derive algebraic form and properties
7. Normalize → **Pearson's coefficient**

8. Extend to continuous case → **Joint Gaussian RVs**

If you want, I can convert this directly into a **LaTeX scribe document (same format as your previous one)**—just tell me 👍